

Лабораторная работа № 14

Статическая маршрутизация в Интернете. Настройка

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Информация

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Последовательность выполнения работы

Настройка линка между
площадками

Настройка интерфейсов коммутатора provider-sw-1

```

Password:

provider-gspolyakov-sw-1>ena
provider-gspolyakov-sw-1>enable
Password:
provider-gspolyakov-sw-1#conf
provider-gspolyakov-sw-1#configure ter
provider-gspolyakov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
provider-gspolyakov-sw-1(config)#int
provider-gspolyakov-sw-1(config)#interface f0/3
provider-gspolyakov-sw-1(config-if)#swit
provider-gspolyakov-sw-1(config-if)#switchport mo
provider-gspolyakov-sw-1(config-if)#switchport mode t
provider-gspolyakov-sw-1(config-if)#switchport mode trunk

provider-gspolyakov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
ec
provider-gspolyakov-sw-1(config-if)#ex
provider-gspolyakov-sw-1(config-if)#exit
provider-gspolyakov-sw-1(config)#interface f0/4
provider-gspolyakov-sw-1(config-if)#switchport mode trunk
provider-gspolyakov-sw-1(config-if)#exit
provider-gspolyakov-sw-1(config)#vlan 5
provider-gspolyakov-sw-1(config-vlan)# name q42
provider-gspolyakov-sw-1(config-vlan)#exit
provider-gspolyakov-sw-1(config)#interface vlan5
provider-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up
no shutdown
provider-gspolyakov-sw-1(config-if)#exit
provider-gspolyakov-sw-1(config)#vlan 6
provider-gspolyakov-sw-1(config-vlan)#name sochi
provider-gspolyakov-sw-1(config-vlan)#exit
provider-gspolyakov-sw-1(config)#interface vlan6
provider-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
no sho
provider-gspolyakov-sw-1(config-if)#no shot
provider-gspolyakov-sw-1(config-if)#no shu
provider-gspolyakov-sw-1(config-if)#no shutdown
provider-gspolyakov-sw-1(config-if)#exit
provider-gspolyakov-sw-1(config)#

```

Настройка интерфейсов маршрутизатора msk-donskaya-gw-1

```
msk-donskaya-gspolyakov-gw-1>ena
msk-donskaya-gspolyakov-gw-1>enable
Password:
msk-donskaya-gspolyakov-gw-1#int
msk-donskaya-gspolyakov-gw-1#inte
msk-donskaya-gspolyakov-gw-1#conf
msk-donskaya-gspolyakov-gw-1#configure tr
msk-donskaya-gspolyakov-gw-1#configure t
msk-donskaya-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-gspolyakov-gw-1(config)#int
msk-donskaya-gspolyakov-gw-1(config)#interface f0/1.5
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
msk-donskaya-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 5
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.255.1 255.255.255.252
msk-donskaya-gspolyakov-gw-1(config-subif)#descri
msk-donskaya-gspolyakov-gw-1(config-subif)#description sochi
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#exi
```



Настройка интерфейсов маршрутизатора msk-q42-gw-1

```
msk-q42-gspolyakov-gw-1>enable
Password:
msk-q42-gspolyakov-gw-1#conf
msk-q42-gspolyakov-gw-1#configure ter
msk-q42-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-gspolyakov-gw-1(config)#int
msk-q42-gspolyakov-gw-1(config)#interface f0/1
msk-q42-gspolyakov-gw-1(config-if)#no shutdown
msk-q42-gspolyakov-gw-1(config-if)#exit
msk-q42-gspolyakov-gw-1(config)#interface f0/1.5
msk-q42-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
msk-q42-gspolyakov-gw-1(config-subif)#encap
msk-q42-gspolyakov-gw-1(config-subif)#enc
msk-q42-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 5
msk-q42-gspolyakov-gw-1(config-subif)#ip address 10.128.255.2 255.255.255.252
% 10.128.255.0 overlaps with FastEthernet0/0
msk-q42-gspolyakov-gw-1(config-subif)#desc
msk-q42-gspolyakov-gw-1(config-subif)#description dons
msk-q42-gspolyakov-gw-1(config-subif)#description donsk
msk-q42-gspolyakov-gw-1(config-subif)#description donskaya
msk-q42-gspolyakov-gw-1(config-subif)#exit
msk-q42-gspolyakov-gw-1(config)#exit
msk-q42-gspolyakov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```


Настройка интерфейсов коммутатора sch-sochi-sw-1

Password:

```
sch-sochi-gspolyakov-sw-1>enable
```

Password:

```
sch-sochi-gspolyakov-sw-1#conf
```

```
sch-sochi-gspolyakov-sw-1#configure ter
```

```
sch-sochi-gspolyakov-sw-1#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
sch-sochi-gspolyakov-sw-1(config)#int
```

```
sch-sochi-gspolyakov-sw-1(config)#interface f0/24
```

```
sch-sochi-gspolyakov-sw-1(config-if)#swit
```

```
sch-sochi-gspolyakov-sw-1(config-if)#switchport mode trunk
```

```
sch-sochi-gspolyakov-sw-1(config-if)#exit
```

```
sch-sochi-gspolyakov-sw-1(config)#interface f0/23
```

```
sch-sochi-gspolyakov-sw-1(config-if)#
```

```
sch-sochi-gspolyakov-sw-1(config-if)#switchport mode trunk
```

```
sch-sochi-gspolyakov-sw-1(config-if)#exit
```

```
sch-sochi-gspolyakov-sw-1(config)#vlan 6
```

```
sch-sochi-gspolyakov-sw-1(config-vlan)#name sochi
```

```
sch-sochi-gspolyakov-sw-1(config-vlan)#exit
```

```
sch-sochi-gspolyakov-sw-1(config)#interface vlan6
```

```
sch-sochi-gspolyakov-sw-1(config-if)#
```

```
%LINK-5-CHANGED: Interface Vlan6, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
```

```
no shutdown
```

```
sch-sochi-gspolyakov-sw-1(config-if)#
```



Настройка интерфейсов маршрутизатора sch-sochi-gw-1

Password:

```
sch-sochi-gspolyakov-gw-1>enable
```

Password:

```
sch-sochi-gspolyakov-gw-1#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
sch-sochi-gspolyakov-gw-1(config)#inter
```

```
sch-sochi-gspolyakov-gw-1(config)#interface f0/0
```

```
sch-sochi-gspolyakov-gw-1(config-if)#no shu
```

```
sch-sochi-gspolyakov-gw-1(config-if)#no shutdown
```

```
sch-sochi-gspolyakov-gw-1(config-if)#exit
```

```
sch-sochi-gspolyakov-gw-1(config)#interface f0/0.6
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/0.6, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up  
enca
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 6
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#ip addre
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
```

```
% 10.128.255.4 overlaps with FastEthernet0/0
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#desc
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#description dons
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#description donskaya
```

```
sch-sochi-gspolyakov-gw-1(config-subif)#exit
```

```
sch-sochi-gspolyakov-gw-1(config)#exit
```

```
sch-sochi-gspolyakov-gw-1#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
|
```

Настройка площадки 42-го квартала

Настройка интерфейсов маршрутизатора msk-q42-gw-1

```
sch-sochi-gspolyakov-gw-1>enable
Password:
sch-sochi-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gspolyakov-gw-1(config)#inter
sch-sochi-gspolyakov-gw-1(config)#interface f0/0
sch-sochi-gspolyakov-gw-1(config-if)#no shu
sch-sochi-gspolyakov-gw-1(config-if)#no shutdown
sch-sochi-gspolyakov-gw-1(config-if)#exit
sch-sochi-gspolyakov-gw-1(config)#interface f0/0.6
sch-sochi-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up
enca
sch-sochi-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 6
sch-sochi-gspolyakov-gw-1(config-subif)#ip addre
sch-sochi-gspolyakov-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
% 10.128.255.4 overlaps with FastEthernet0/0
sch-sochi-gspolyakov-gw-1(config-subif)#desc
sch-sochi-gspolyakov-gw-1(config-subif)#description dons
sch-sochi-gspolyakov-gw-1(config-subif)#description donskaya
sch-sochi-gspolyakov-gw-1(config-subif)#exit
sch-sochi-gspolyakov-gw-1(config)#exit
sch-sochi-gspolyakov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
enable
sch-sochi-gspolyakov-gw-1#configure ter
sch-sochi-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gspolyakov-gw-1(config)#inter
sch-sochi-gspolyakov-gw-1(config)#interface f0/0
sch-sochi-gspolyakov-gw-1(config-if)#no shut
sch-sochi-gspolyakov-gw-1(config-if)#no shutdown
sch-sochi-gspolyakov-gw-1(config-if)#exit
sch-sochi-gspolyakov-gw-1(config)#interface f0/0.6
sch-sochi-gspolyakov-gw-1(config-subif)#enca
sch-sochi-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 6
sch-sochi-gspolyakov-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
% 10.128.255.4 overlaps with FastEthernet0/0
sch-sochi-gspolyakov-gw-1(config-subif)#description donskaya
sch-sochi-gspolyakov-gw-1(config-subif)#e
```

Настройка интерфейсов коммутатора msk-q42-sw-1

```
msk-q42-gspolyakov-gw-1#configur
msk-q42-gspolyakov-gw-1#configure te
msk-q42-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-gspolyakov-gw-1(config)#inter
msk-q42-gspolyakov-gw-1(config)#interface f0/0 no shutd
msk-q42-gspolyakov-gw-1(config)#interface f0/0 no shutdo
msk-q42-gspolyakov-gw-1(config)#interface f0/0 no shutdown
^
% Invalid input detected at '^' marker.

msk-q42-gspolyakov-gw-1(config)#interface f0/0
msk-q42-gspolyakov-gw-1(config-if)#no sh
msk-q42-gspolyakov-gw-1(config-if)#no shutdown
msk-q42-gspolyakov-gw-1(config-if)#exit
msk-q42-gspolyakov-gw-1(config)#exit
msk-q42-gspolyakov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-gspolyakov-gw-1(config)#interface f0/0.201
msk-q42-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up
enc
msk-q42-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 201
msk-q42-gspolyakov-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
% 10.129.0.0 overlaps with FastEthernet0/0
msk-q42-gspolyakov-gw-1(config-subif)#description q42-main
msk-q42-gspolyakov-gw-1(config-subif)#exit
msk-q42-gspolyakov-gw-1(config)#interface f1/0
msk-q42-gspolyakov-gw-1(config-if)#no shutdown
msk-q42-gspolyakov-gw-1(config-if)#exit
msk-q42-gspolyakov-gw-1(config)#interface f1/0.202
msk-q42-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet1/0.202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

msk-q42-gspolyakov-gw-1(config-subif)#enca
msk-q42-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 202
msk-q42-gspolyakov-gw-1(config-subif)#ip add
msk-q42-gspolyakov-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
% 10.129.1.0 overlaps with FastEthernet0/0
msk-q42-gspolyakov-gw-1(config-subif)#description q42-management
```

Настройка интерфейсов маршрутизирующего коммутатора

```
msk-q42-gspolyakov-sw-1>enable
Password:
msk-q42-gspolyakov-sw-1#confu
msk-q42-gspolyakov-sw-1#confi
msk-q42-gspolyakov-sw-1#configure ter
msk-q42-gspolyakov-sw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-gspolyakov-sw-1(config)#interf
msk-q42-gspolyakov-sw-1(config)#interface f0/24
msk-q42-gspolyakov-sw-1(config-if)#swit
msk-q42-gspolyakov-sw-1(config-if)#switchport mode trunk
msk-q42-gspolyakov-sw-1(config-if)#exit
msk-q42-gspolyakov-sw-1(config)#interf
msk-q42-gspolyakov-sw-1(config)#interface f0/1
msk-q42-gspolyakov-sw-1(config-if)#swit
msk-q42-gspolyakov-sw-1(config-if)#switchport mode acc
msk-q42-gspolyakov-sw-1(config-if)#switchport mode access
msk-q42-gspolyakov-sw-1(config-if)#swi
msk-q42-gspolyakov-sw-1(config-if)#switchport acc
msk-q42-gspolyakov-sw-1(config-if)#switchport access vl
msk-q42-gspolyakov-sw-1(config-if)#switchport access vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-gspolyakov-sw-1(config-if)#exit
msk-q42-gspolyakov-sw-1(config)#vlan 201
msk-q42-gspolyakov-sw-1(config-vlan)#name q42-main
msk-q42-gspolyakov-sw-1(config-vlan)#exit
msk-q42-gspolyakov-sw-1(config)#inter
msk-q42-gspolyakov-sw-1(config)#interface vlan201
msk-q42-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up
no shu
msk-q42-gspolyakov-sw-1(config-if)#no shutdown
msk-q42-gspolyakov-sw-1(config-if)#exit
msk-q42-gspolyakov-sw-1(config)#
```

Настройка интерфейсов коммутатора msk-hostel-sw-1

```
msk-hostel-gspolyakov-gw-1#  
%SYS-5-CONFIG_I: Configured from console by console  
  
msk-hostel-gspolyakov-gw-1#con  
msk-hostel-gspolyakov-gw-1#con  
msk-hostel-gspolyakov-gw-1#conf  
msk-hostel-gspolyakov-gw-1#configure t  
msk-hostel-gspolyakov-gw-1#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
msk-hostel-gspolyakov-gw-1(config)#int  
msk-hostel-gspolyakov-gw-1(config)#interface g0/1  
msk-hostel-gspolyakov-gw-1(config-if)#sw  
msk-hostel-gspolyakov-gw-1(config-if)#sw  
msk-hostel-gspolyakov-gw-1(config-if)#sw  
msk-hostel-gspolyakov-gw-1(config-if)#sw  
msk-hostel-gspolyakov-gw-1(config-if)#switchport trunk encapsulation dot1q  
^  
% Invalid input detected at '^' marker.  
  
msk-hostel-gspolyakov-gw-1(config-if)#switchport mode trunk  
^  
% Invalid input detected at '^' marker.  
  
msk-hostel-gspolyakov-gw-1(config-if)#exit  
msk-hostel-gspolyakov-gw-1(config)#interface f0/1  
%Invalid interface type and number  
msk-hostel-gspolyakov-gw-1(config)#
```

Настройка площадки в Сочи

Настройка интерфейсов маршрутизатора sch-sochi-gw-1

```
msk-hostel-gspolyakov-sw-1>enab
msk-hostel-gspolyakov-sw-1>enable
Password:
msk-hostel-gspolyakov-sw-1#confi
msk-hostel-gspolyakov-sw-1#configure ter
msk-hostel-gspolyakov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-gspolyakov-sw-1(config)#inter
msk-hostel-gspolyakov-sw-1(config)#interface g0/1
%Invalid interface type and number
msk-hostel-gspolyakov-sw-1(config)#interface g0/1
%Invalid interface type and number
msk-hostel-gspolyakov-sw-1(config)#interface f0/1
msk-hostel-gspolyakov-sw-1(config-if)#sw
msk-hostel-gspolyakov-sw-1(config-if)#switchport mode access
msk-hostel-gspolyakov-sw-1(config-if)#switchport access vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-gspolyakov-sw-1(config-if)#exit
msk-hostel-gspolyakov-sw-1(config)#vlan 301
msk-hostel-gspolyakov-sw-1(config-vlan)#name hostel-main
msk-hostel-gspolyakov-sw-1(config-vlan)#no shutdown
      ^
% Invalid input detected at '^' marker.

msk-hostel-gspolyakov-sw-1(config-vlan)#exit
msk-hostel-gspolyakov-sw-1(config)#inte
msk-hostel-gspolyakov-sw-1(config)#interface vlan301
msk-hostel-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
interface vlan301
msk-hostel-gspolyakov-sw-1(config-if)#no shutdown
msk-hostel-gspolyakov-sw-1(config-if)#exit
msk-hostel-gspolyakov-sw-1(config)#
```

Настройка интерфейсов коммутатора sch-sochi-sw-1

```
sch-sochi-gspolyakov-gw-1>enable
Password:
sch-sochi-gspolyakov-gw-1#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gspolyakov-gw-1(config)#interface f0/0.401
sch-sochi-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up
enca
sch-sochi-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 401
sch-sochi-gspolyakov-gw-1(config-subif)#ip address 10.131.0.1 255.255.255.0
% 10.131.0.0 overlaps with FastEthernet0/0
sch-sochi-gspolyakov-gw-1(config-subif)#des
sch-sochi-gspolyakov-gw-1(config-subif)#description sochi-main
sch-sochi-gspolyakov-gw-1(config-subif)#exit
sch-sochi-gspolyakov-gw-1(config)#inter
sch-sochi-gspolyakov-gw-1(config)#interface f0/0.402 ip address 10.130.1.1 255.255.255.0
^
% Invalid input detected at '^' marker.

sch-sochi-gspolyakov-gw-1(config)#interface f0/0.402
sch-sochi-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.402, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up

sch-sochi-gspolyakov-gw-1(config-subif)#enca
sch-sochi-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 402
sch-sochi-gspolyakov-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
% 10.130.1.0 overlaps with FastEthernet0/0
sch-sochi-gspolyakov-gw-1(config-subif)#description sochi-managem
```

Настройка маршрутизации между площадками

```
sch-sochi-gspolyakov-sw-1>enable
Password:
sch-sochi-gspolyakov-sw-1#enab
sch-sochi-gspolyakov-sw-1#enable t
sch-sochi-gspolyakov-sw-1#enable t
sch-sochi-gspolyakov-sw-1#con
sch-sochi-gspolyakov-sw-1#conf
sch-sochi-gspolyakov-sw-1#configure ter
sch-sochi-gspolyakov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gspolyakov-sw-1(config)#interface f0/1
sch-sochi-gspolyakov-sw-1(config-if)#sw
sch-sochi-gspolyakov-sw-1(config-if)#switchport mo
sch-sochi-gspolyakov-sw-1(config-if)#switchport mode acc
sch-sochi-gspolyakov-sw-1(config-if)#switchport mode access
sch-sochi-gspolyakov-sw-1(config-if)%%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/1 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsistent port type.

sch-sochi-gspolyakov-sw-1(config-if)#swi
sch-sochi-gspolyakov-sw-1(config-if)#switchport acc
sch-sochi-gspolyakov-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-gspolyakov-sw-1(config-if)#exit
sch-sochi-gspolyakov-sw-1(config)#vlan 401
sch-sochi-gspolyakov-sw-1(config-vlan)#name sochi
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (401), with provider-gspolyakov-sw-1 FastEtherne
sch-sochi-gspolyakov-sw-1(config-vlan)#name sochi-main
sch-sochi-gspolyakov-sw-1(config-vlan)#exit
sch-sochi-gspolyakov-sw-1(config)#inter
sch-sochi-gspolyakov-sw-1(config)#interface vlan401
sch-sochi-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

sch-sochi-gspolyakov-sw-1(config-if)#no shut
sch-sochi-gspolyakov-sw-1(config-if)#no shutdown
sch-sochi-gspolyakov-sw-1(config-if)#exit
sch-sochi-gspolyakov-sw-1(config)#
```

Настройка маршрутизатора msk-donskaya-gw-1

```
msk-donskaya-gspolyakov-gw-1>con
msk-donskaya-gspolyakov-gw-1>ena
msk-donskaya-gspolyakov-gw-1>enable
Password:
msk-donskaya-gspolyakov-gw-1#conf
msk-donskaya-gspolyakov-gw-1#configure ter
msk-donskaya-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-gspolyakov-gw-1(config)#ip ro
msk-donskaya-gspolyakov-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-gspolyakov-gw-1(config)#
```

Настройка маршрутизатора msk-q42-gw-1

```
msk-q42-gspolyakov-gw-1>con
msk-q42-gspolyakov-gw-1>ena
msk-q42-gspolyakov-gw-1>enable
Password:
msk-q42-gspolyakov-gw-1#con
msk-q42-gspolyakov-gw-1#conf
msk-q42-gspolyakov-gw-1#configure ter
msk-q42-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-gspolyakov-gw-1(config)#ip rou
msk-q42-gspolyakov-gw-1(config)#ip rout
msk-q42-gspolyakov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-gspolyakov-gw-1(config)#
```

```
sch-sochi-gspolyakov-gw-1 (config-sub1) # exit  
sch-sochi-gspolyakov-gw-1 (config) #  
sch-sochi-gspolyakov-gw-1 (config) # ip route 0.0.0.0 0.0.0.0 10.128.255.5  
sch-sochi-gspolyakov-gw-1 (config) #
```

Настройка маршрутизации на 42 квартале

Настройка маршрутизатора msk-q42-gw-1

```
msk-hostel-gspolyakov-gw-1>ena
msk-hostel-gspolyakov-gw-1>enable
Password:
msk-hostel-gspolyakov-gw-1#conf
msk-hostel-gspolyakov-gw-1#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-gspolyakov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.2
%Invalid next hop address (it's this router)
msk-hostel-gspolyakov-gw-1(config)#
```


Настройка интерфейсов маршрутизирующего коммутатора

```
msk-donskaya-gspolyakov-gw-1>con
msk-donskaya-gspolyakov-gw-1>ena
msk-donskaya-gspolyakov-gw-1>enable
Password:
msk-donskaya-gspolyakov-gw-1#conf
msk-donskaya-gspolyakov-gw-1#configure ter
msk-donskaya-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-gspolyakov-gw-1(config)#ip ro
msk-donskaya-gspolyakov-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-gspolyakov-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.6
msk-donskaya-gspolyakov-gw-1(config)#
msk-donskaya-gspolyakov-gw-1(config)#int
msk-donskaya-gspolyakov-gw-1(config)#interface f0/1.5
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nap ins
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nap inside
      ^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat ins
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat inside
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#ip acc
msk-donskaya-gspolyakov-gw-1(config)#ip access-list ex
msk-donskaya-gspolyakov-gw-1(config)#ip access-list extended li
msk-donskaya-gspolyakov-gw-1(config)#ip access-list extended nas
msk-donskaya-gspolyakov-gw-1(config)#ip access-list extended nat
msk-donskaya-gspolyakov-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#remark q42
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit ip host 10.130.128.200 any
```

Настройка NAT на маршрутизаторе msk-donskaya-gw-1
